

Dexter Fernandes

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https://github.com/Dexter-Fernandes

Senior Machine Learning Engineer

Machine Learning Engineer with 3+ years delivering production-grade computer vision systems across safety-critical industrial settings, including fault-tolerant safety vision stacks for Tier-1 clients. Experienced in leading projects until they get shipped to production. Skilled in synthetic data generation, adversarial-condition training, and hypothesis-driven experimentation to improve robustness, generalisation, and reliability on edge and cloud platforms.

WORK EXPERIENCE

Esbaaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Built predictive models for classifying 24/7 oil-rig operations using time-series and CV features, achieving 23% higher F1 and supporting automated risk triage.
- Developed real-time perception modules for oil-rig safety systems, deploying CNN-based vision pipelines with YOLOv5 on edge devices to improve detection accuracy by 15%.
- Implemented automated data ingestion, cleaning, training, and validation pipelines for large multimodal sensor datasets using MLflow, DVC, and Optuna, streamlining experimentation and reducing model iteration time by 40%.
- Built synthetic image generation and augmentation procedures to improve model robustness under harsh industrial conditions, reducing false positives and supporting long-term operational reliability.
- Led on-site validation and diagnostics on oil-rig deployments, conducting root-cause analysis of model failures and improving overall system stability under environmental stressors.
- Implemented continuous integration and testing workflows for edge inference modules, improving maintainability and deployment speed across hardware targets.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Developed adversarial-condition training and validation suites simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Built and maintained large-scale image and sensor datasets, automating labelling, versioning, and performance tracking to support anomaly investigation and continuous improvement cycles.
- Repurposed legacy object detection models for automated labelling pipeline workflows, reducing manual annotation effort by 80% and accelerating dataset curation.
- Mentored engineers on optimisation and deployment practices, raising code quality and delivery consistency.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

CERTIFICATIONS

Deep Learning Specialisation

deeplearning.ai

Jul 2019

PROJECTS

End-to-End Damage Severity & Claim Assessment

- Developed an end-to-end automotive claims assessment pipeline combining computer vision-based damage detection with a trained severity-classification model, achieving accurate triage for real-world vehicle damage scenarios.
- Built a lightweight repair-insight engine that transforms model outputs into insurer-friendly explanations, using structured reasoning + LLM-generated narratives to deliver clear, interpretable claim summaries.
- Designed the entire workflow (data preprocessing, model training, evaluation, and a user-facing inference module) demonstrating practical understanding of insurance claims processes.

SKILLS

- **Deep Learning and Optimisation:** Bayesian Optimisation, DeepSORT, MultiRocket, ONNX, OpenVINO, Optuna, PyTorch, Quantisation-Aware Training, TensorFlow, TensorRT
- **Perception & SLAM:** Convolutional Neural Networks, NumPy, OpenCV, Vision Transformer (ViT)
- **Cloud, Edge & MLOps:** Apache Kafka, AWS, Docker, DVC (Data Version Control), FastAPI, Git, Intel VPUs, Linux, Microsoft Azure, MLFlow, NVIDIA Jetson
- **Programming & Development:** Bash, C++, Pandas, PostgreSQL, Python